WELCOME!

Online Training Course

Foundations in Mobile Phone Data (MPD) for Policy:
Design and implementation for MPD Initiatives

Data Quality and Characteristics

Session 5

Overall Course overview

- Five online training sessions:
  - Introduction to Planning an MPD Initiative
  - Policy Applications
  - Developing Partnerships and Negotiating Data Access
  - Data Pipeline Architecture
  - Data Quality and Characteristics
- Two self-paced topics:
  - Data governance and safeguards
  - Communicating in MPD initiatives

Housekeeping information

- Delivery timeslot for online training sessions
    - Every Wednesday this September
    - From 12-1pm UTC

- The slides and a session recording will be uploaded to the Open Learning Campus after delivery, for your reference
- Additional resources, quizzes and/or other follow-up will also be made available after sessions
- In-house translation from English to French in-session; slides and tools/resource documents will be provided afterwards in French, Spanish and Portuguese
- Try to keep Q&A until the end

Data quality & characteristics

Module 1 - Session 1.5

Overview

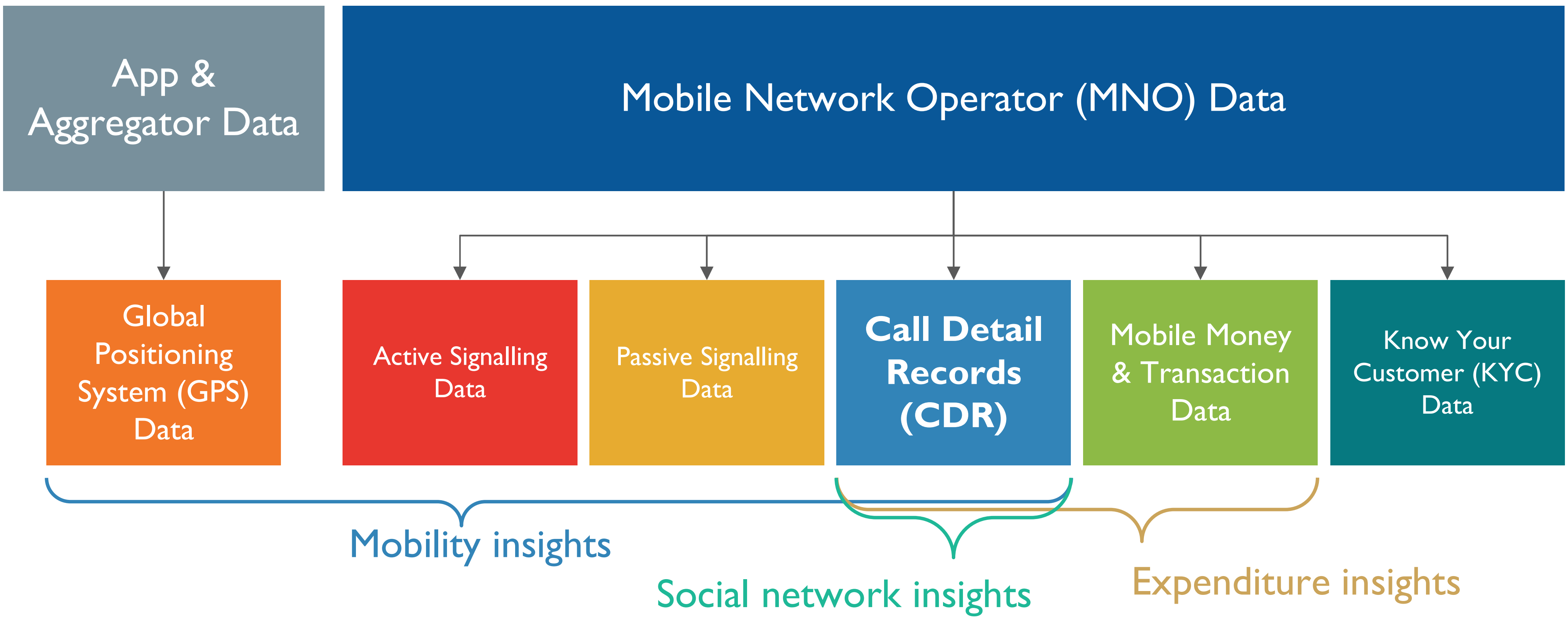
- Introduction to MPD data
  - Key information included in CDR Data
  - Strengths and limitations of CDR Data
- Quality considerations in using CDR data for policy purposes
  - Impact of CDR data characteristics on statistical quality
  - Adjusting biases in statistics derived from CDR Data
- Quality assurance framework

Learning objectives

- Describe what CDR data is and how it is generated
- Describe key CDR data characteristics for producing statistics
- List some of the strengths and limitations of CDR data
- Describe some of the factors that can affect the characteristics of statistics produced from CDR data
- Describe the data sources or surveys used for adjusting biases, how this can be done, and some of the pitfalls or challenges involved
- Describe frameworks for quality assurance in statistical production from CDR data

Introduction to Mobile Phone Data

Mobile Phone Data (MPD)



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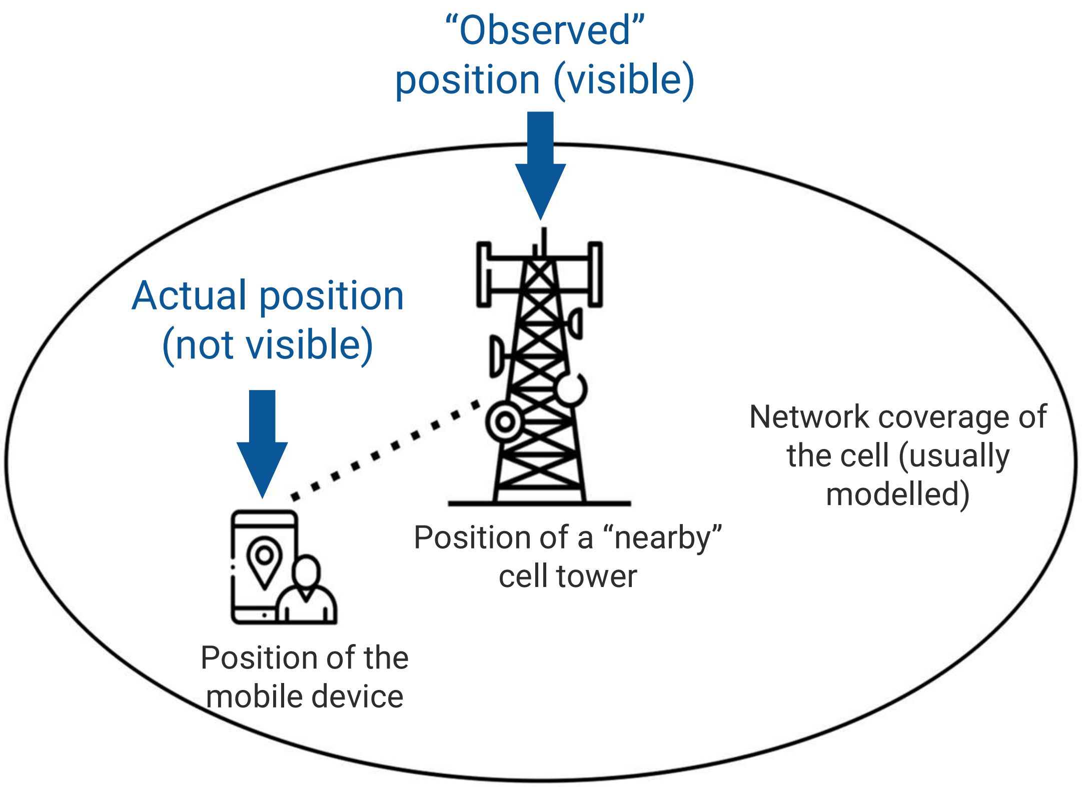
MPD - GPS data vs CDR data

	Data collected /stored by	Spatial precision & resolution	Temporal resolution	Population coverage
CDR data	Mobile Network Operator	Varies across areas Approximate Cell tower location	Intermittent Depend on user behavior	All phones
GPS data	Smartphone app	Accurate Exact Device location	Continuous Frequent & constant	App/OS users of smartphones

How CDR data is generated

CDR data

- Passively generated when a subscriber
  - Makes or receives **a call**
  - Sends or receives **an SMS**
  - Uses mobile **data**
- Location is at **the cell tower** level
- Routinely stored by MNOs for billing purposes



Key information contained in CDR data

MSISDN	CELL_ID	TIMESTAMP
AA204V1542DCA00	451154211	2016-10-10 15:35:25
AA20[illegible]	[illegible]312	20[illegible]3:45
AA204V[illegible]	[illegible]312	20[illegible]1:56
AA204V1542DCA03	451354312	2016-10-10 21:59:32
AA204V1542DCA04	452616792	2016-10-10 22:42:23
B45QHV45CAEVA5	476126941	2016-10-10 08:13:21
B45QHV45CAEVA6	476126941	2016-10-10 08:14:15
B45QHV45CAEVA7	47612[illegible]	[illegible]016-10-10 08:14:59
B45QHV45CAEVA8	47612[illegible]	[illegible]016-10-10 12:41:01
B45QHV45CAEVA9	476126941	2016-10-10 13:10:45
B45QHV45CAEVA10	476126941	2016-10-10 15:20:43

Calling party identifier (pseudonymised)

Timestamp

Cell ID (location)

Other variables in CDR data

- Receiving party identifier (pseudonymised)
- Network event type

CDR data do **NOT** contain information about the contents of a network event

Pseudonymization

- Pseudonymization means removing personally identifiable information by replacing it with random strings.
- The same original value always gets the same strings after pseudonymized

Example:
A phone number in mobile data is replaced with a random string like "A1B26s9Lrjd8UK7g45NC3".

Attribute	Original value	algorithm	Encrypted value
Phone No	654003453	SHA256	7cdab53309c015854b433f0f34d6cbc015104f9c42e36539b35ef974b26122d9
IMSI	611050105799949	SHA256	5c0c2c3675e41471d746fbe721e435e9ac4cdcba3406d78be89e5a87cab2d5f9
IMEI	865760021379230	SHA256	1341a37ec42db25140af0c6f7cc6e28b2b45f5d66832f6337fd6995e0d1f1582

Identifiers of CDR data

	MSISDN Mobile Station International Subscriber Directory Number	**IMSI** International Mobile Subscriber Identity	**IMEI** International Mobile Equipment Identity
Represents	• **Phone number** • Used for billing and routing calls/SMS	• Unique to the **SIM card** • Identifies the subscriber within the mobile network.	• Unique to the **device** • identifies the physical handset
Assigned To	SIM card / subscriber	SIM card	Mobile device

Strengths and limitations

Strengths

- High penetration globally
- Large geographic scale
- High quality data
- Near-real time
- High granularity
- Automatically generated
- Already generated by organisations

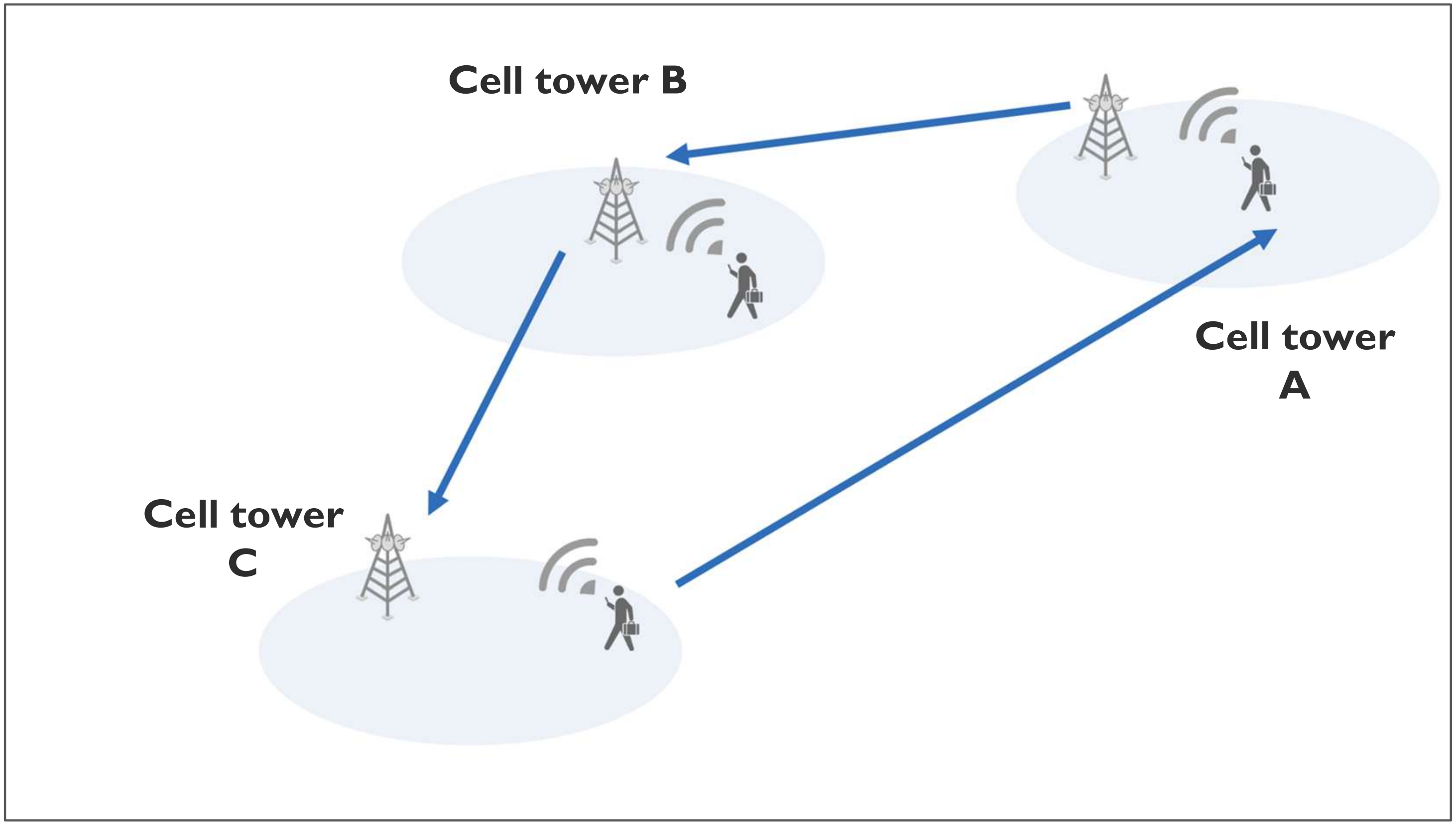
Limitations

- Precision limited by cell tower density
- Representativeness of mobile phone subscriptions
- One subscription may not represent one individual, and vice versa
- Large datasets requiring suitable data infrastructure

Quality considerations in using CDR data for policy purposes

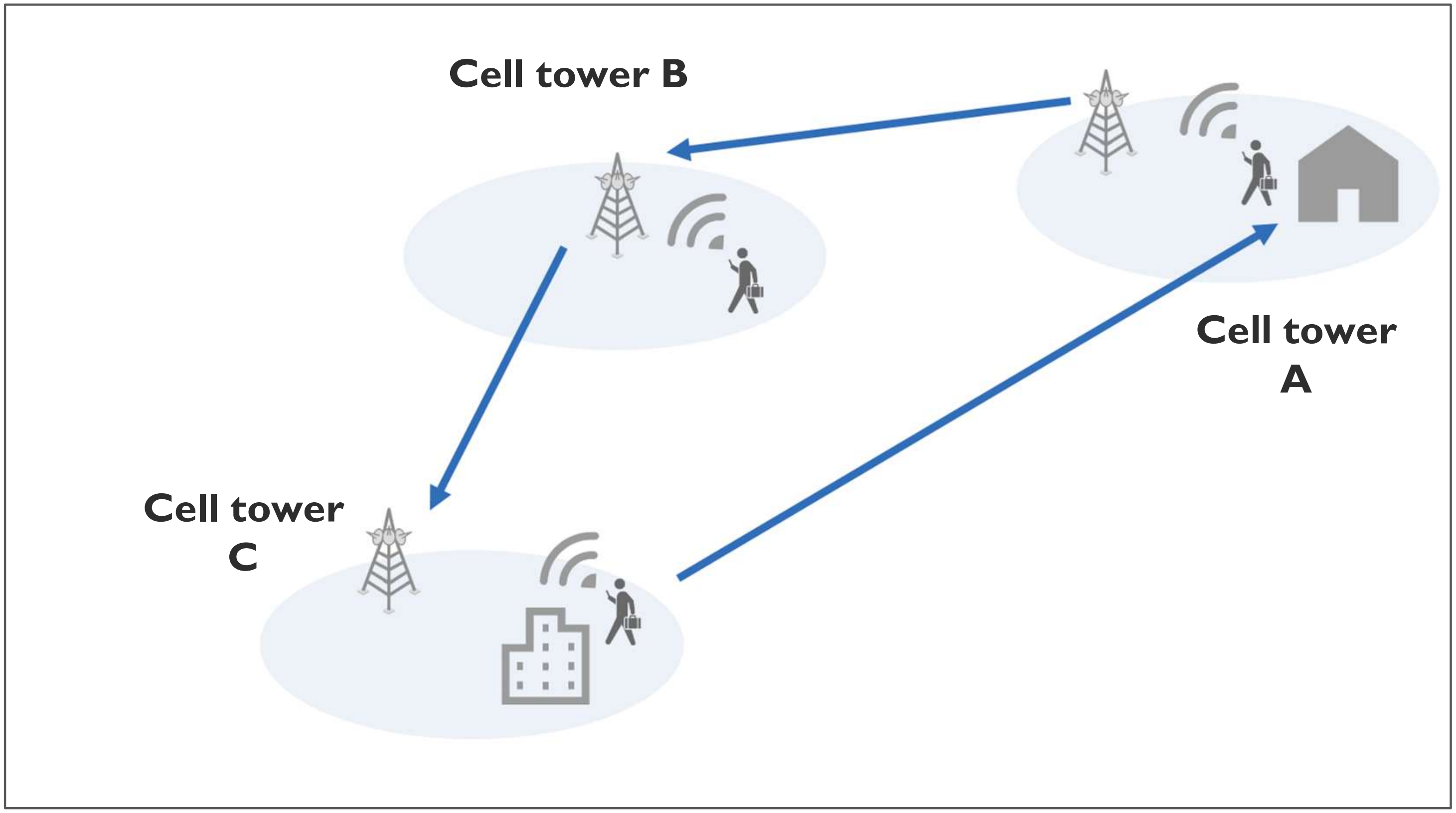
Device location is observed as cell tower location

Time	Cell Tower
6.00	A
6.35	A
7.25	B
8.00	B
9.40	C
10.45	C
13.00	C
14.50	C
16.00	C
17.30	C



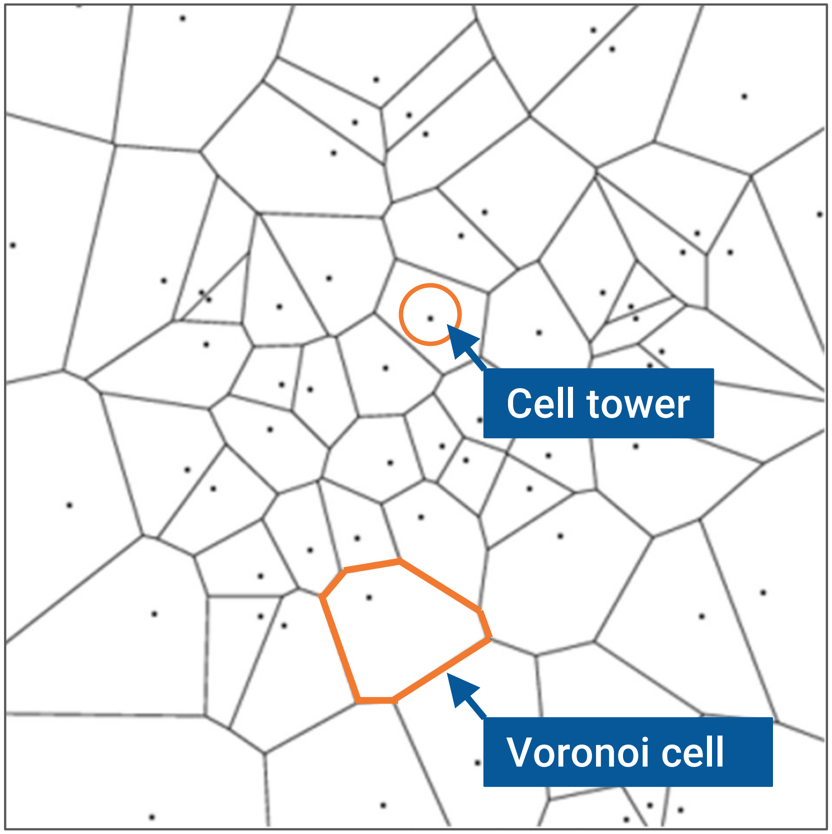
Timestamps are used to infer the type of locations

Time	Cell Tower
6.00	A
6.35	A
7.25	B
8.00	B
9.40	C
10.45	C
13.00	C
14.50	C
16.00	C
17.30	C

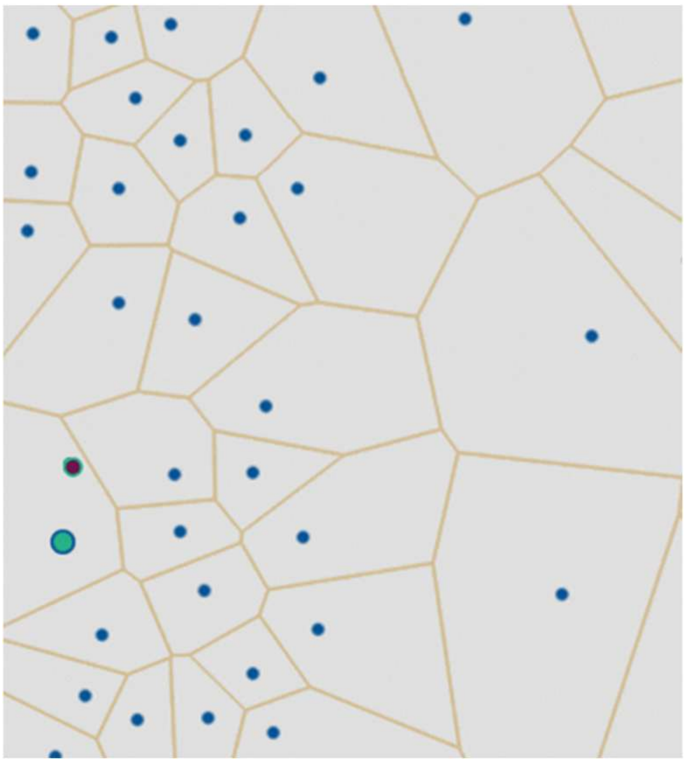


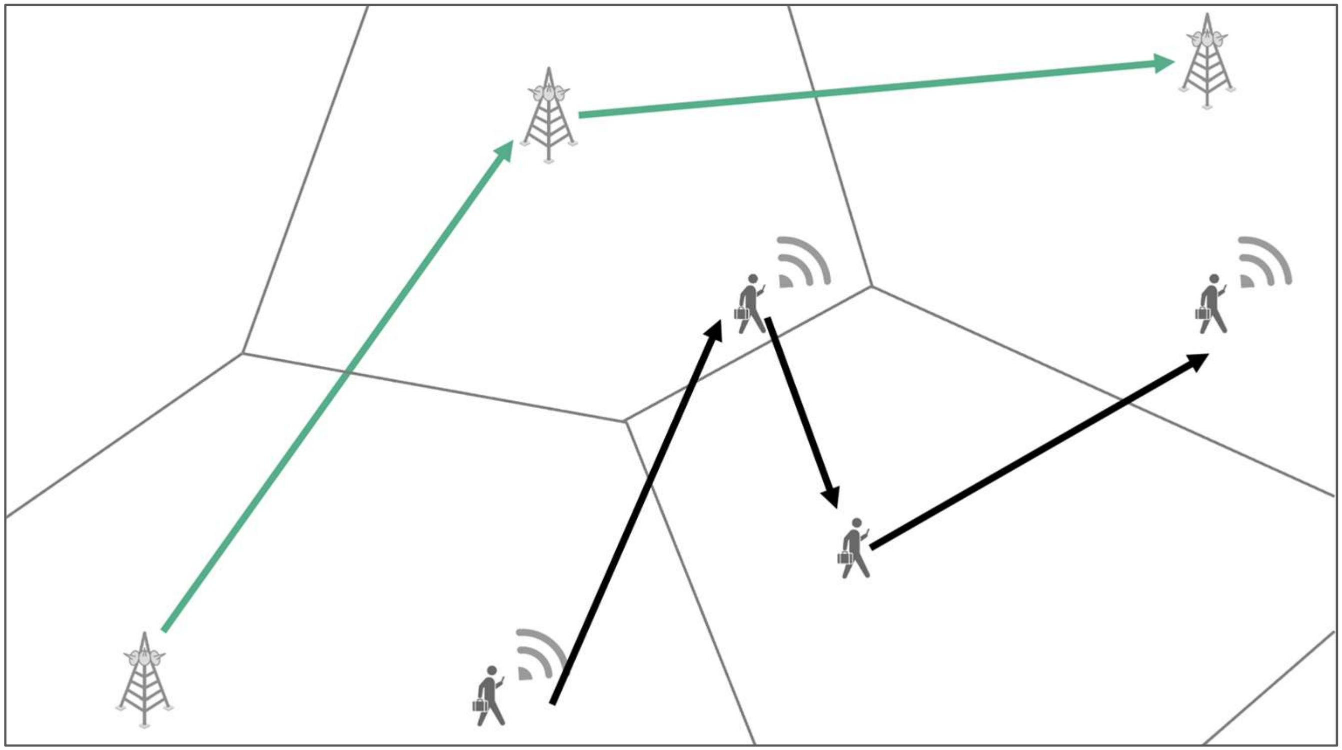
Hypothetical area represented by each cell tower: Voronoi cells

- A Voronoi diagram based on cell tower locations **divides a given area into regions associated with the nearest cell tower**.
- Each region (voronoi cell) in the diagram **represents the hypothetical area that a cell tower network covers**



Flows among cell towers can deviate from actual mobility





Subscriber movements
Network events
Observed trajectory
Cell tower coverage
Cell towers

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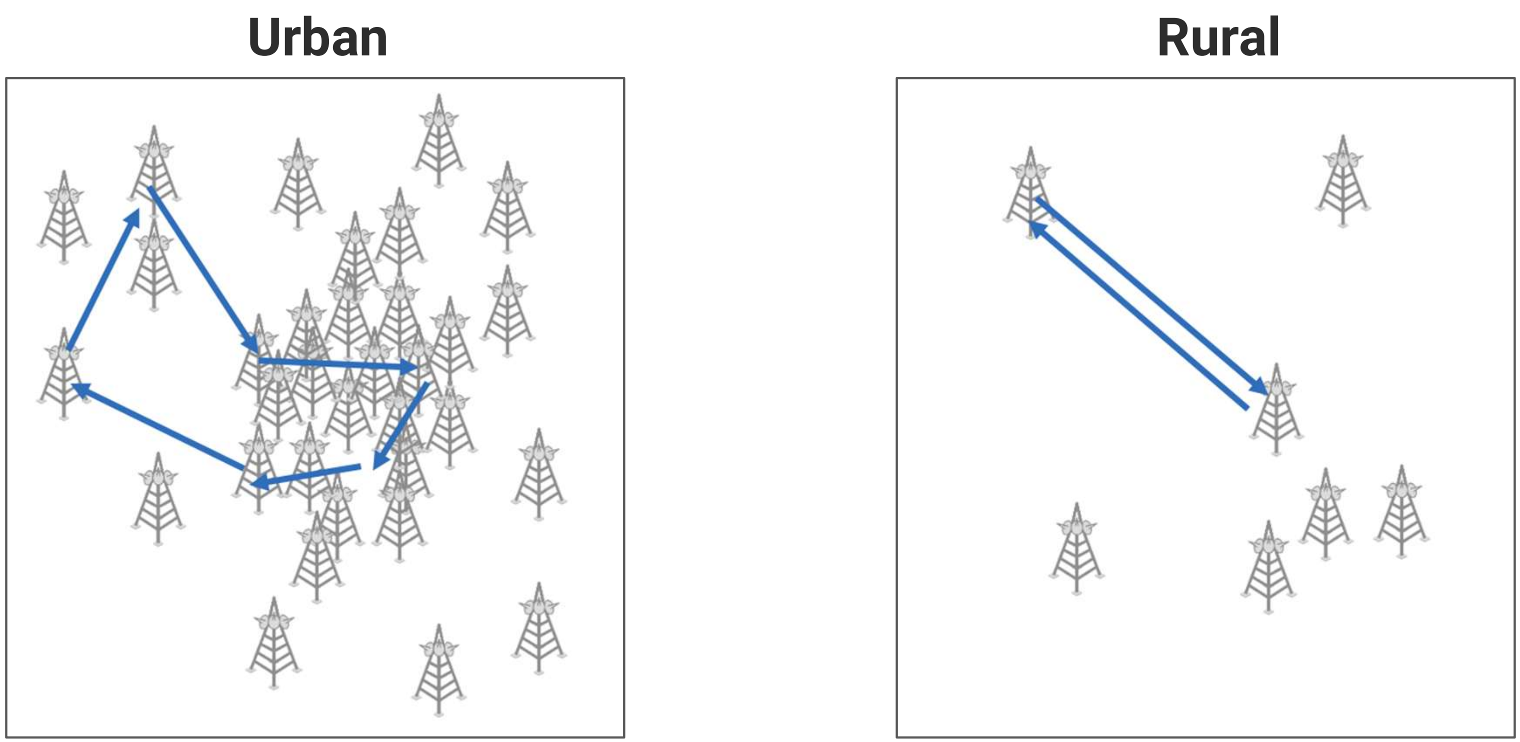
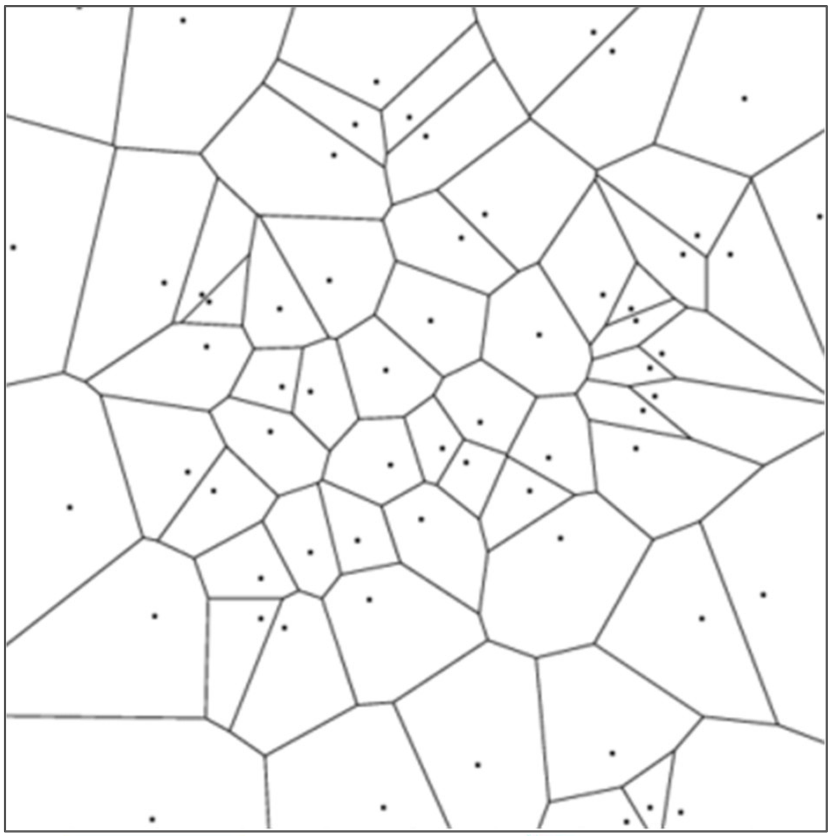
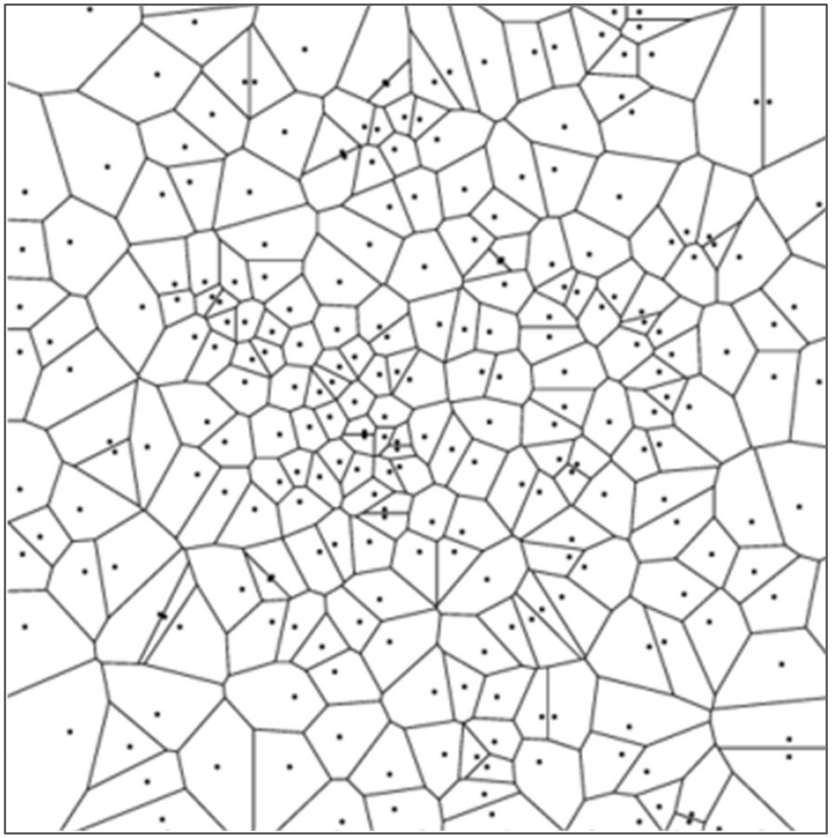
Cell tower density can limit the mobility observable depending on localities



More cell towers allows for greater spatial precision





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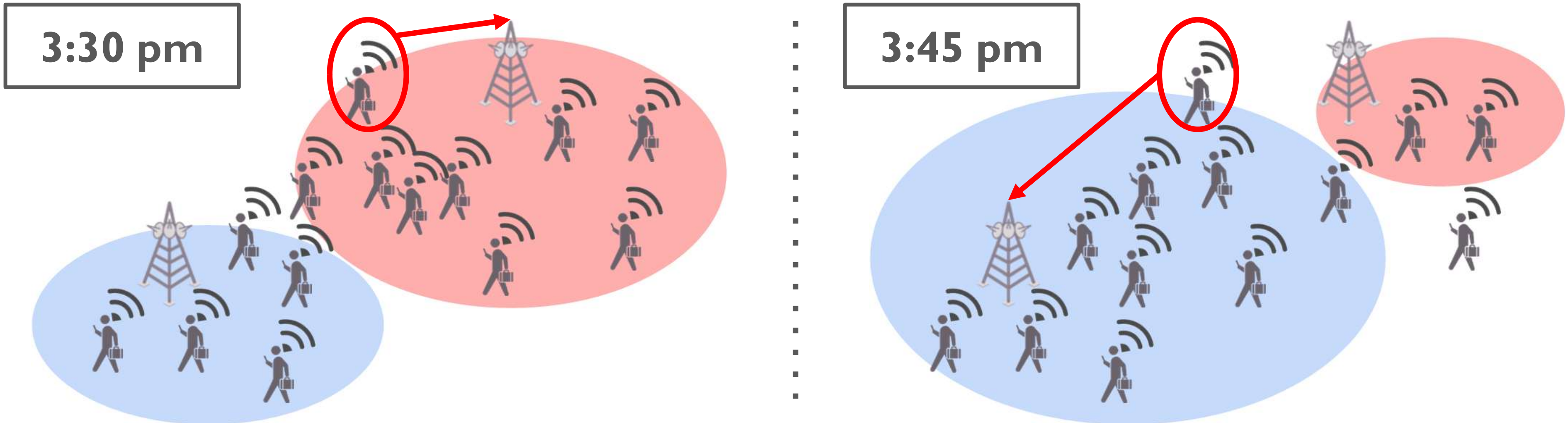
Frequent tower switching (handover) causes location noise

Tower switching occurs due to signal strength changes, load balancing, and technology switch by tower.



Usage habits affect the estimation accuracy

①

Time	Cell Tower
6.00	A
6.35	A
7.25	B
8.00	B
9.40	C
10.45	C
13.00	C
14.50	C

②

Time	Cell Tower
6.00	A
6.35	A
7.25	B
8.00	B
9.40	C
17.30	C
20.20	A
22.50	

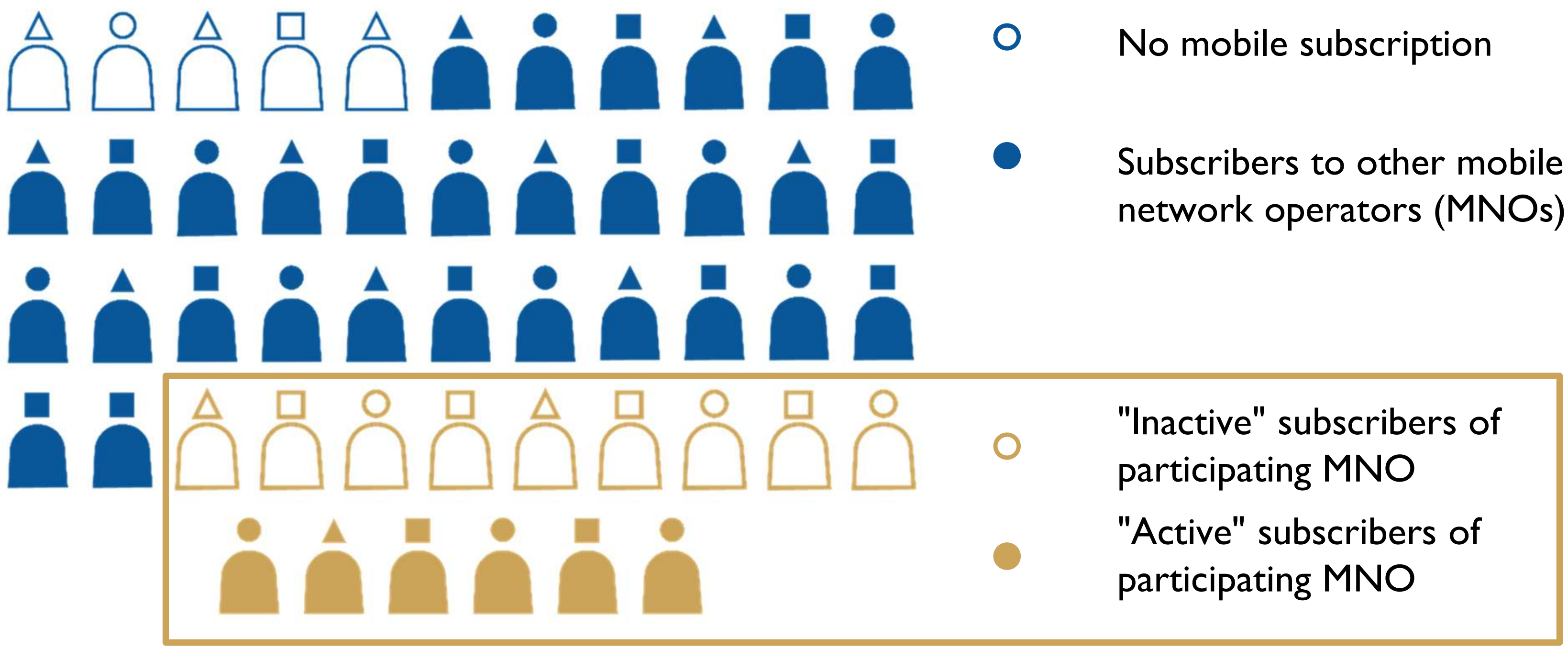
③

Time	Cell Tower
6.00	A
6.35	A
20.20	A
20:50	A
21:35	A
22.50	A

Usage habits can vary by…

Wealthier/poorer, age, gender, occupation, urban/rural (connectivity to the network), multiple-sim holding, phone sharing, etc.

Who is in the CDR data you are accessing?



Visual: Flowminder

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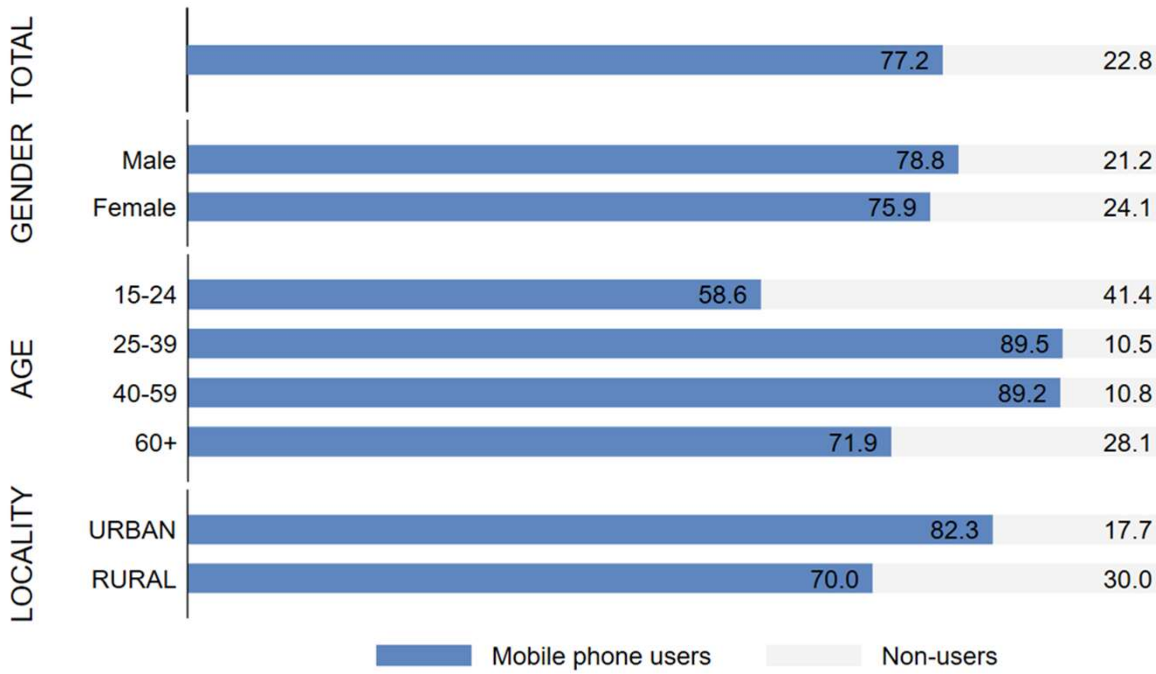
Mobile phone use by demographic | Ghana

Mobile phone use is:

- More common among **men** than among women
- More common among **persons of working-age** than among the elderly, and among children
- More widespread in **urban areas** than in rural areas



Source: AHIES 2022, Q3, weighted(pop_weight) (Flowminder)

Issue of bias

A whole population
you are trying to estimate

Visual: Flowminder

Method for population-weighted adjustment

Data

CDR data, Census data (or existing population estimates such as gridded datasets), additional primary/secondary survey data

Method

1. Get initial estimate of the residential population for each cell tower and aggregate at the district level using CDR data
2. Compute the scaling factor for each district by dividing the census population by the CDR estimated population
3. Adjust the CDR-derived estimates by multiplying them by the scaling factors

Make further adjustment based on additional data availability (phone users & non-users, MNO subscription, multiple-SIM holding)

Examples of additional data for adjustment

Census: Gambia Bureau of Statistics (GBoS) included questions about ICT facility and device in the last Census conducted in 2023/24 (supported by the World Bank)

- Access to the mobile device
    - Mobile phone users or non-users
- Usage of the mobile services
    - Multiple-sim holding, phone sharing
    - MNO subscriptions
    - Expenditure on mobile phone services

Household Travel Survey

- Typical travel behavior on a weekday
- Home and work/education locations on the transport analysis zone (TAZ)

Data sources per country

Data sources	DRC	Ghana	Haiti
CDRs	Vodacom	Telecel (formerly Vodafone) Ghana	Digicel
Baseline population	WorldPop 2020: gridded population estimates (bottom-up)	Census 2021: sub-regional population estimates (GSS)	WorldPop, IHSI, UN Population Prospects: sub-regional population estimates
Secondary & "piggyback" surveys	Multi-Sector Needs Assessment (MSNA) 2023: update to weighting parameters (REACH)	Annual Household Income and Expenditures Survey (AHIES) 2022: phone users and non-users (GSS)	Multi-Sector Needs Assessment (MSNA) survey 2022: phone users and non-users (REACH)
Primary surveys	Micro-census 2021: covering 7 provinces, phone users and non-users (HH survey, led by Flowminder) Phone survey 2021: targeting phone users across the country from all MNOs (commissioned by Flowminder)	Phone survey 2022: targeting phone users across the country from all MNOs (commissioned by Flowminder, conducted by GSS)	Phone survey 2023: targeting phone users across the country from all MNOs (commissioned by Flowminder, conducted by Viamo)

Source: Flowminder

Quality assurance framework

Guiding principles to maintain trust

Guiding principles to maintain public trust in the use of mobile operator data for policy purposes

Data & Policy (2021), 3: e24
doi:10.1017/dap.2021.21

CAMBRIDGE UNIVERSITY PRESS

TRANSLATIONAL ARTICLE

Guiding principles to maintain public trust in the use of mobile operator data for policy purposes

Ronald Jansen[1]*, Karoly Kovacs[1], Siim Esko[2], Erki Saluveer[2], Kaja Sõstra[3], Linus Bengtsson[4], Tracey Li[4], Wole A. Adewole[4], Jade Nester[5], Ayumi Arai[6] and Esperanza Magpantay[7]

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Five Principles to ensure trust

- Necessity and proportionality (Fit-for-purpose)
- Professional independence
- Privacy protection
- Commitment to quality
- International comparability

Trade-off and balancing

- Necessity and proportionality (Fit-for-purpose) <-> Quality
- Professional independence
- Privacy protection <-> Quality <-> Fit-for-purpose
- Commitment to quality <-> Privacy protection
- International comparability

Quality Assurance Framework of the European Statistics System (QAF)

- Provides good practice, methods, tools at
  - Institutional environment level,
  - Process level, and
  - Output level.
- Collection of good practices
- Adopted in May 2019

https://ec.europa.eu/eurostat/documents/64157/4392716/ESS-QAF-V1-2final.pdf/bbf5970c-1adf-46c8-afc3-58ce177a0646

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Sector-Specific Quality Assurance Processes

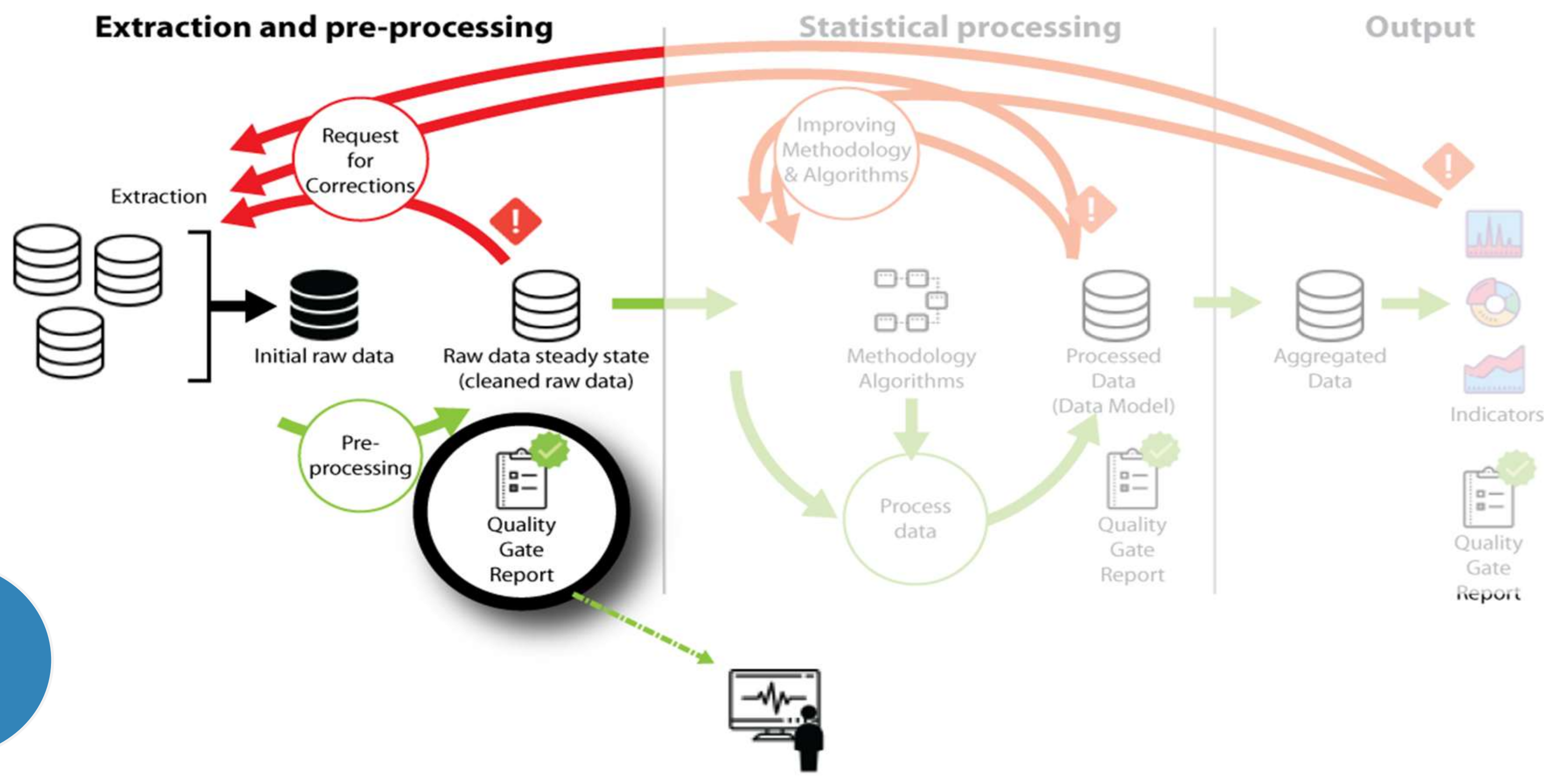
Processing steps

	Input	Throughput	Output
Source	Privacy and security		Confidentiality
Metadata	Log files Metadata Consistency …	System independence Quality gates Steady states	Accessibility and clarity Relevance
Data	Consistency Validity …		Coherence Consistency Validity …

Based on: A Suggested Framework for the Quality of Big Data – UNECE
https://statswiki.unece.org/download/attachments/108102944/Big%20Data%20Quality%20Framework%20-%20final-%20Jan08-2015.pdf

UNECE Big Data Quality Task Team

Quality Gate I – Input Data



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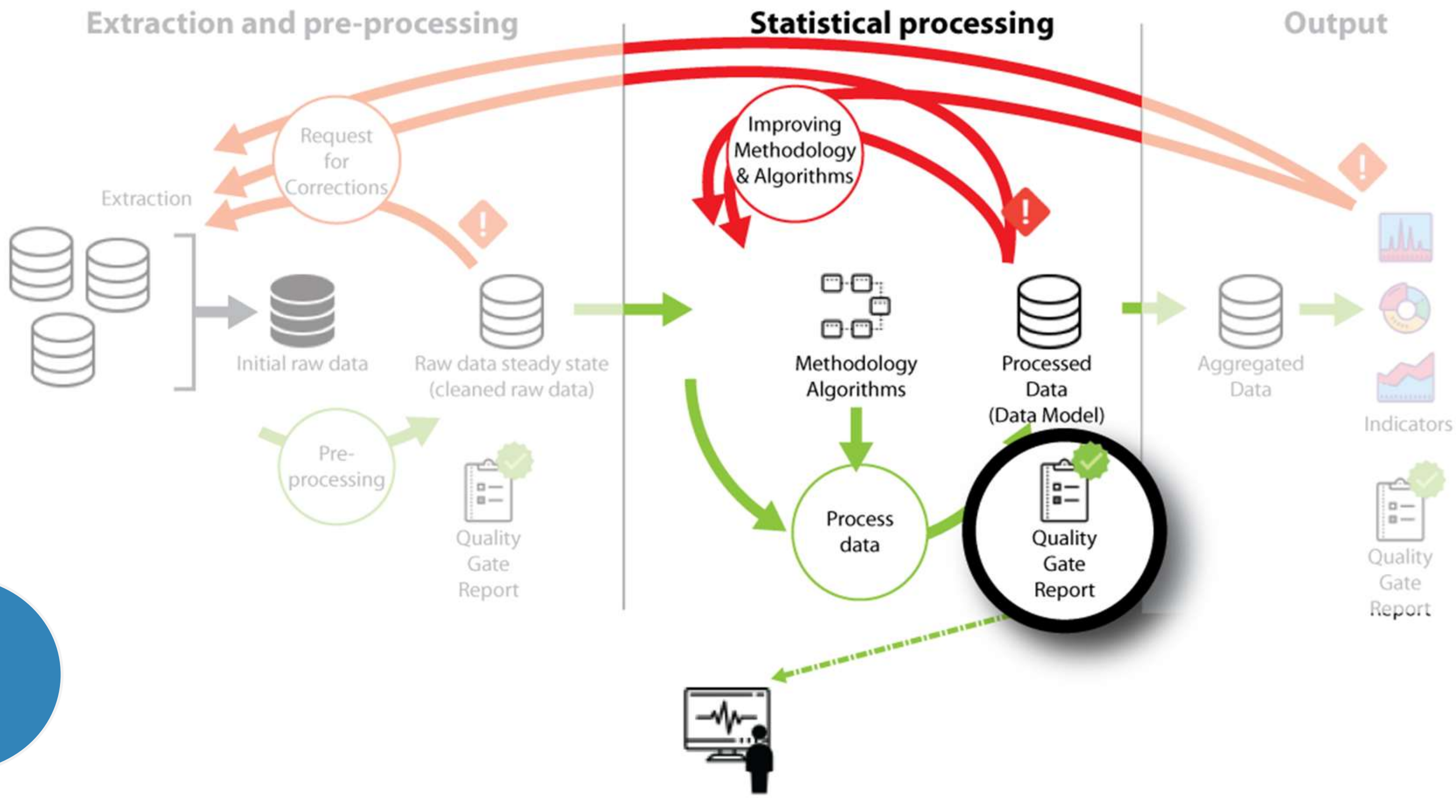
Quality Gate 2 – Processing



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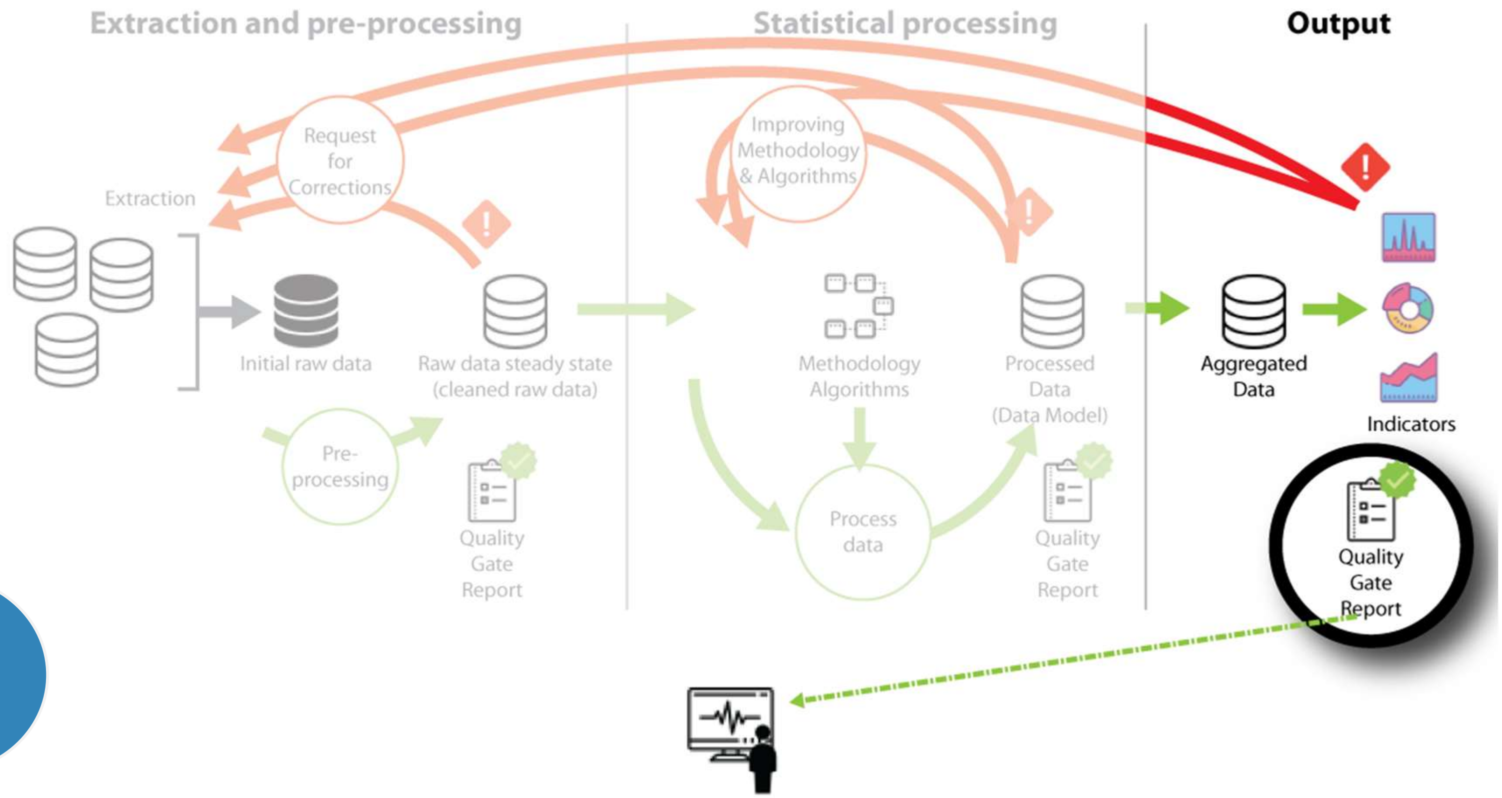
Quality Gate 3 – Output Data



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Quality Gates in summary

Quality Gate 1: Input data	Ensures raw data are consistent, logically sound, and valid for statistical use, with errors corrected by mobile operators before moving on.
Quality Gate 2: Processing	Focuses on system design and processing, where parameters are tested, logs reviewed, and methods adjusted until the system works properly.
Quality Gate 3: Output data	Checks final results for accuracy, consistency, and comparability, validating them against reference data where possible.

Thank you!

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